

EGH 404 Assessment 2 – Data Analysis

You should draw across the unit content for Data Analysis—data wrangling, data visualisation, statistics, correlation and regression – to answer this assessment. Please provide your answers directly on this template. Do not delete the template wording as this assists markers in knowing which question you are answering. You have the option of completing this assessment either using MATLAB, Python or Excel.

Remember to provide evidence to justify your responses. Evidence includes but is not limited to any forms of visualisation, source code (for MATLAB or Python users), screenshots of Excel formulas, and screenshots of Excel “Data Analysis” tools. Please include your supporting evidence at the end of the report as an appendix (copy and paste the code into the document, or if using live scripts, merge the document PDF with the exported live script PDF). Please note that your code will not be run except under exceptional circumstances, but may be used to evaluate the process that you used to answer the question. Figures should be well formatted and include clear titles, axis labels, legends (where applicable), and appropriate units.

SECTION 1 – PROCESSING AND VISUALISATION OF REAL-WORLD DATA

Brisbane City Council (BCC) is considering upgrades to a number of bikeway networks. To do so, they need to understand how different bikeways are being used by both cyclists and pedestrians, so they can use this information to plan suitable bikeway upgrades. They are using data they have gathered from sensors placed along the bikeway paths, which automatically record the daily number of cyclists and pedestrians.

BCC has requested that you conduct an initial assessment regarding cyclist *and* pedestrian usage of two specific bikeway paths - Bicentennial Bikeway in Brisbane and North Brisbane Bikeway in Windsor. BCC is investigating historical bikeway usage of these two bikeway paths from 2014-2018 (5 years), and if there has been any relationship between different weather factors (rainfall, solar exposure and maximum temperature) and the use of these bikeway paths by both cyclists and pedestrians.

We have provided you with the BCC Bikeway Data for 2014-2018 (between 01 January 2014 and 31 December 2018) for daily cyclist and pedestrian usage of two locations: Bicentennial Bikeway in Brisbane and North Brisbane Bikeway in Windsor. Furthermore, Bureau of Meteorology (BOM) weather data captured at Brisbane City’s weather station for the same period above for three weather factors: daily rainfall, solar exposure and maximum temperatures is also provided.

Please refer to the Assessment page in Canvas for a copy of this dataset and scripts to load the data (only for Python and MATLAB users)

Download the brisbane_bikeway_counters.csv and brisbane_weather.csv datasets.

S1.T1: Bikeway Data Inspection

- a) Create a **single** box plot visualisation (separate box plots for each bikeway in a SINGLE visualisation) to help you identify outliers across daily bikeway usage data. Your box plot should include box data for:
 - Cyclist usage on Bicentennial Bikeway, Brisbane

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- Pedestrian usage on Bicentennial Bikeway, Brisbane
- Cyclist usage on North Brisbane Bikeway, Windsor
- Pedestrian usage on North Brisbane Bikeway, Windsor

(1 pts)

Hint: Module on - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

(Add your answer here)

- b) Create a **single** line graph visualisation to help identify any errors, missing or other problematic data across daily bikeway usage data. Your line graph visualisation should include 4 separate lines to represent usage over time for:
- Cyclist usage on Bicentennial Bikeway, Brisbane
 - Pedestrian usage on Bicentennial Bikeway, Brisbane
 - Cyclist usage on North Brisbane Bikeway, Windsor
 - Pedestrian usage on North Brisbane Bikeway, Windsor

(1 pts)

Hint: Module on - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

(Add your answer here)

- c) Based on your answers in (a) and (b) above, have you identified any problematic data (i.e. outliers, missing data, errors, etc.) in your bikeways data? If yes, please identify the problems you have found, the dates (or date ranges if the identified problem persists for a consecutive period) of the problematic data points, and how you intend to handle the problematic data in your final (cleaned) dataset, with justification for the method of handling. Provide your answers in the table below (you can add more rows if needed), and for each bikeway and problem type. Provide **up to 3 examples for each problem type** (e.g., outliers, missing data, errors) for **each bikeway**. i.e. If more than 3 instances exist → report only 3; If fewer than 3 exist → report all that you can find.

(4 pts)

Hint: Module on - Introduction to Visualising Data & Statistics could be beneficial as you develop your response.

Name of the column with the problematic	Specify the date or date range (for continuing	How do you intend to handle the	Justification
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data, type of the problem (outliers, missing data, errors) and the value.	missing values) with problematic data	problematic data identified?	

S1.T2: Weather Data Inspection

- a) Create **separate boxplots** for each weather factor to help identify outliers for:

- daily rainfall
- daily maximum temperatures
- daily solar exposure

(1 pts)

Hint: Module on - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

(Add your answer here)

- b) Create **separate line graphs** to help identify any errors, missing or other problematic data across time for each of the following weather factors

- daily rainfall
- daily maximum temperatures
- daily solar exposure

(1 pts)

Hint: Module on - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

(Add your answer here)

- d) Based on your answers in (a) and (b) above, have you identified any problematic data (i.e. outliers, missing data, errors, etc.) in your weather data? If yes, please identify the problems you have found, the dates (or date ranges if the identified problem persists for a consecutive period) of the problematic data points, and how you intend to handle the problematic data in your final (cleaned) dataset, with justification for the method of handling. Provide your answers in the table below (you can add more rows if needed), and for each bikeway and problem type. Provide **up to 3 examples for each problem type** (e.g., outliers, missing data, errors) for **each bikeway**. i.e. If more than 3 instances exist → report only 3; If fewer than 3 exist → report all that you can find.

(4 pts)

Name of the column with the problematic data, type of the problem (outliers, missing data, errors) and the value.	Specify the date or date range (for continuing missing values) with problematic data	How do you intend to handle the problematic data identified?	Justification

SECTION 2 – CORRELATION ANALYSIS USING REAL-WORLD DATA

Using your cleaned data, please answer the following questions.

BCC wants to further investigate the relationship between daily cyclist and pedestrian usage on the Bicentennial Bikeway in Brisbane and the North Brisbane Bikeway in Windsor for 2014-2018. To answer Sections 2 and 3, please use the cleaned datasets (brisbane_bikeway_counters_cleaned.csv, brisbane_weather_cleaned.csv) provided in Canvas Assessment 2 – Details page.

Please answer the following questions:

S2.T1: Correlation Coefficient

Please compute the correlation coefficients for each of the relationships listed in the table below.

Hint: Module on - Correlation Analysis along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

(2 pts)

Please compute the correlation coefficient between:	Correlation coefficient
(a) Cyclists and Pedestrians on the Bicentennial Bikeway in Brisbane	
(b) Cyclists and Pedestrians on the North Brisbane Bikeway in Windsor	
(c) Cyclists on Bicentennial Bikeway in Brisbane and Cyclists on the North Brisbane Bikeway in Windsor	
(d) Pedestrians on Bicentennial Bikeway in Brisbane and Cyclists on the North Brisbane Bikeway in Windsor	

S2.T2: Correlation Visualisation

Please create a scatter plot to visualise the relationship between daily cyclist usage of North Brisbane Bikeway in Windsor (x-axis) and daily pedestrian usage (y-axis) for the Bicentennial Bikeway in Brisbane.

(2 pts)

Hint: Module on - Correlation Analysis along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

(Add your visualisation here)

S2.T3: Correlation Analysis Interpretation (400 words)

Using the correlation coefficient calculated in 2.1(a) and the visualisation produced in 2.2 above, what is your interpretation of the relationship between cyclist usage of North Brisbane Bikeway in Windsor and pedestrian usage on Bicentennial Bikeway in Brisbane between 2014-2018? Please use your visualisation, correlation coefficient outputs and data patterns over time to support your interpretation of bikeway usage.

(4 pts)

Hint: Module on - Correlation Analysis could be beneficial as you develop your response.

(Add your answer here)

SECTION 3 – MULTIPLE LINEAR REGRESSION ANALYSIS

Using your cleaned data, please answer the following questions.

To answer Sections 2 and 3, please use the cleaned datasets (brisbane_bikeway_counters_cleaned.csv, brisbane_weather_cleaned.csv) provided in Canvas Assessment 2 – Details page.

BCC wants to further investigate how daily weather factors (i.e. daily rainfall, solar exposure, maximum temperature) influence usage of the Bicentennial Bikeway, Brisbane – and whether there is any difference in how weather influences cyclist versus pedestrian usage of this bikeway. To answer the questions in this section, please split the dataset as follows: use data from January 1st,

2014, to December 31st, 2017 (inclusive) as the training set, and use all remaining data as the testing set.

S3.T1: Multiple Regression Models & Supporting Visualisations

Please conduct Multiple Regression analyses to investigate if one or more daily weather factors (i.e. rainfall, solar exposure, maximum temperature) influence **pedestrian usage on the Bicentennial Bikeway, Brisbane**. Do not create separate regression models for each independent variable. Instead, build a single regression model that includes all the identified independent variables. You may need to repeat the regression modelling process until all variables have statistically significant p-values.

- Provide model summary table (including P-values, R^2 , adjusted R^2 , RMSE on train and test sets)
- Generate scatter plots illustrating the observed data points for each independent variable against the dependent variable and overlay the fitted regression line to demonstrate the model's representation of the relationship.
- Discuss Any model simplification (i.e. removal of independent variables from the model) with appropriate justification. Discuss the impact of simplifying the model on performance if any.

(3 pts)

Hint: Module on - Regression Analysis along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

(Add your multiple regression model outputs & visualisations here)

Please conduct Multiple Regression analyses to investigate if one or more daily weather factors (i.e. rainfall, solar exposure, maximum temperature) influence **cyclist usage on the Bicentennial Bikeway, Brisbane**. Provide a model summary table (including P-values, R^2 , adjusted R^2 , RMSE on the test-set). You may need to repeat the regression modelling process until all variables have statistically significant p-values.

- Provide model summary table (including P-values, R^2 , adjusted R^2 , RMSE on train and test sets)
- Generate scatter plots illustrating the observed data points for each independent variable against the dependent variable and overlay the fitted regression line to demonstrate the model's representation of the relationship.
- Discuss Any model simplification (i.e. removal of independent variables from the model) with appropriate justification. Discuss the impact of simplifying the model on performance if any.

(3 pts)

Hint: Module on - Regression Analysis along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

(Add your multiple regression model outputs & visualisations here)

S3.T2: Multiple Regression Interpretation (400 words)

What is your interpretation of the results produced from the two Multiple Regression Models above, and why? Specifically, provide:

- (a) Predicted vs actual values plot on the test-set and discuss the predictive power of the regression models in S3.T1 (a) and (b).
- (b) Discuss the suitability and the applicability of the model using statistical measures (e.g. regression coefficients, R^2 , adjusted R^2 , RMSE, and p-values).
- (c) Discuss the reasons for any identified performance gap between training and testing, supported by appropriate plots that you have generated above.

(5 pts)

Hint: Module on - Regression Analysis could be beneficial as you develop your response.

(Add your answer here)

SECTION 4 – FINDING REAL-WORLD DATA OR SYNTHESISING DATA FOR PRELIMINARY RESEARCH ANALYSIS

In This Section, Choose One Pathway (S4.TA or S4. TB) — Do Not Complete Both

S4.TA: Locating Real-World Dataset (Publicly Available)

You will locate and justify a publicly available real-world tabular dataset that is broadly related to your research topic from Assessment 1. Note that you will use the same dataset during your Assessment 3 – Project Pitch presentation, to demonstrate preliminary evaluation.

Choose a dataset that meets all of the following:

1. Tabular format (e.g., CSV, Excel, TSV).
2. Contains ≥ 100 rows (recommended) and ≥ 5 variables/columns, with at least 3 quantitative variables.
3. Is publicly accessible and has a clear source link and licensing/usage statement (or terms of use).
4. Is broadly related to your research topic area (it does not need to match your topic exactly). You must explain how it supports preliminary analysis relevant to your topic.

Examples of places to find datasets:

- UCI Machine Learning Repository: archive.ics.uci.edu
- U.S. Government open data portal: data.gov
- Australian Government open data portal: data.gov.au
- NASA open data catalogue: data.nasa.gov

Write a short justification ($\approx 150 - 250$ words) explaining why this dataset is useful as a preliminary proxy for your chosen research topic.

Using the following table, describe your dataset

Variable	Min	Max	Units
Var 1			
Var 2			

S4.TB: Synthetic Data Generation

Using the synthetic data generation procedure introduced during the practical session, create a synthetic dataset that aligns with the research question you identified in your response to Assignment 1. You should use the same research topic from Assessment 1 to define the variables for synthetic data generation and for your Assessment 3 – Project Pitch presentation. Ensure the dataset is relevant, appropriately structured, and suitable for subsequent analysis.

Your synthetic dataset must include:

- Up to four independent variables and one dependent variable (i.e., 3–5 key variables total).
- At least 200 rows.
- Clear definitions of:
 - variable ranges (min/max), units (where meaningful), and distribution shapes (e.g., normal, uniform, skewed).
 - any assumptions about relationships (linear/nonlinear/none).

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Write a short justification ($\approx 150 - 250$ words) explaining why this dataset is useful as a preliminary proxy for your chosen research topic.

Using the following table describe your dataset

Variable	Min	Max	Units	Distribution	Justification and other assumptions
Var 1					
Var 2					

(4 pts)

SECTION 5 – CORRELATION ANALYSIS USING PRELIMINARY DATA

Using the dataset you used in Section 4 (i.e. real-world data you located or the synthetic dataset you generated) answer the following questions.

S5.T1: Data Modelling Using Preliminary Data

- In your dataset, identify two or more variables that appear to be related (i.e., feature(s) and a target). Provide scatter plots and a correlation matrix.
- Describe the nature of their relationship (e.g., linear, inverse, nonlinear, or no clear pattern). Support your answer using outputs generated using relevant data analysis concepts (e.g., correlation coefficient, scatter plots, etc.).
- Compare the relationships you observe in your data with those reported in the relevant literature for your chosen topic.

(5 pts)

(Add your answer here)

APPENDIX

Include supporting evidence here